

Sem-II (ANALYSIS - II)

Ch 29 Mean Value Thm (K.A. Ross)

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Q1 (29.1) $f(x) = \sin x$ on $[0, \pi]$

As f is cont on $[0, \pi]$ & diff on $]0, \pi[$
then $\exists c \in]0, \pi[$ s.t. $f'(c) = \frac{f(\pi) - f(0)}{\pi - 0}$

$$= \frac{\sin \pi - \sin 0}{\pi - 0}$$

$$= 0$$

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$$f'(x) = \cos x = 0$$

$$\Rightarrow x = \pi/2 \in]0, \pi[$$

Q2 (29.1) $f(x) = \text{sgn}(x)$ on $[-2, 2]$

As f is not cont at $x=0$

$\therefore f$ is not cont $[-2, 2]$

$\therefore$ hypothesis of M.V. Thm fail.

Q3 (29.2) prove that $|\cos x - \cos y| \leq |x - y|$
 $\forall x, y \in \mathbb{R}$
(do your self)

Q4 (29.3) Suppose f is diff on $\mathbb{R}$ & that
 $f(0) = 0$, $f(1) = 1$ & $f(2) = 1$

(a) T. Show $f'(c) = \frac{1}{2}$ for some $c \in (0, 2)$

As f is cont on $[0, 2]$ & diff on $]0, 2[$
by M.V. Thm

$$\exists c \in (0, 2) \text{ s.t. } f'(c) = \frac{f(2) - f(0)}{2 - 0}$$

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$$= \frac{1 - 0}{2 - 0} = \frac{1}{2}$$

b) T. Show $f'(c) = \frac{1}{7}$ for some $c \in (0, 2)$

Now as f is cont on $[0, 2]$ & diff on $]0, 2[$ & $f(1) = 1 = f(2)$

then by Rolle's thm

$$\exists c_1 \in]0, 2[\text{ s.t. } f'(c_1) = 0$$

$$\therefore f'(c_1) < \frac{1}{7} < f'(c_2) \text{ on }]c_1, c_2[$$

$$\exists c \in]c_1, c_2[\text{ s.t. } f'(c) = \frac{1}{7}$$

Q5 (29.4) We define a function on $[a, b]$

$$h(x) = f(x) \cdot e^{g(x)}$$

since f & g are diff on I , so f & g are cont on I & $a, b \in I$

$\Rightarrow f$ & g are cont on $[a, b]$, Also

f & g are diff on $]a, b[$

Then by chain Rule $x \rightarrow e^{g(x)}$ is cont on $[a, b]$ & diff on $]a, b[$

$\therefore h(x)$ is cont on $[a, b]$ & diff $]a, b[$

$$\& h(a) = f(a) e^{g(a)} = 0$$

$$h(b) = f(b) e^{g(b)} = 0$$

Then by Rolle's thm $\exists c \in]a, b[$

$$\text{s.t. } h'(c) = 0$$

$$h'(c) = f'(c) e^{g(c)} + f(c) \cdot g'(c) e^{g(c)} = 0$$

$$= e^{g(c)} \{ f'(c) + f(c) g'(c) \} = 0$$

$$e^{g(c)} \neq 0$$

$$\therefore f'(c) + f(c) g'(c) = 0$$

Q6(29.5) $|f(x) - f(y)| \leq (x-y)^2 \quad \forall x, y \in \mathbb{R}$

prove that f is constant function.

Sol. Let $y=c \in \mathbb{R}$

by given condition

$$|f(x) - f(c)| \leq (x-c)^2 \quad \forall x \in \mathbb{R}$$

$$\leq |x-c|^2 \quad \forall x \in \mathbb{R}$$

$$\therefore |f(x) - f(c)| \leq |x-c| |x-c| \quad \forall x \in \mathbb{R}$$

$$\frac{|f(x) - f(c)|}{|x-c|} \leq |x-c| \quad \forall x \in \mathbb{R}$$

$$\lim_{x \rightarrow c} \frac{|f(x) - f(c)|}{|x-c|} \leq \lim_{x \rightarrow c} |x-c| = 0$$

$$|f'(c)| \leq 0 \quad \text{Also } |f'(c)| \geq 0$$

$$\therefore f'(c) = 0$$

clearly f is a constant fn on $\mathbb{R}$

Q7(29.6) (do you self)

Q8(29.7) given: $f''(x) = 0 \quad \forall x \in I$

$$\therefore f'(x) = a \quad \forall x \in I$$

$$\therefore f(x) = ax \quad \forall x \in I$$

$$\text{Let } h(x) = f(x) - ax \quad \forall x \in I$$

$$\text{Let } h(x) = b \text{ (const)} \quad \forall x \in I$$

$$\{ \text{Reason: } h'(x) = f'(x) - a = 0 \quad \therefore f'(x) = a \}$$

$$\therefore b = f(x) - ax \quad \forall x \in I$$

$$\therefore f(x) = ax + b$$

Q9 (29.9) s.t. that $e^x \leq e^x \quad \forall x \in \mathbb{R}$

Let $f(x) = e^x - ex \quad \forall x \in \mathbb{R}$

$\therefore f'(x) = e^x - e \quad \therefore f'(x) = 0$

$e^x = e \quad \therefore \boxed{x=1}$

Now $f'(x) > 0$ if $x > 1$ &

$f'(x) < 0$ if $x < 1$

$\Rightarrow f$ is str $\uparrow$ on $[1, \infty]$ where $x > 1$

f is str $\downarrow$ on $(-\infty, 1]$ " $x < 1$

Since f is $\uparrow$ on $[1, \infty]$ so $f(x) > f(1) \quad \forall x > 1$

& f is $\downarrow$ on $(-\infty, 1]$ so $f(x) > f(1) \quad \forall x < 1$

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$e^x - ex > e - e \Rightarrow e^x - ex > 0$

$\therefore e^x > ex \quad \forall x \neq 1$

Q10 (29.11) If $f'(x) > 0$ then f is $\uparrow$

Q11 (29.11) T. show $x < \tan x \quad \forall x \in]0, \frac{\pi}{2}[$

Let $f(x) = \tan x - x \quad \forall x$

$f'(x) = \sec^2 x - 1 > 0 \quad \forall x \in]0, \frac{\pi}{2}[$

$\therefore f$ is str $\uparrow$ on $]0, \frac{\pi}{2}[$

$\Rightarrow f(x) > f(0) \quad \forall x$

$\Rightarrow \tan x - x > 0$

$\Rightarrow \tan x > x$

Q12 (29.12) (b) $f(x) = \frac{x}{\sin x}$ str $\uparrow$ on $]0, \pi[$

$$f'(x) = \frac{\cos x (1 - \tan x) - x}{\sin^2 x} \quad \text{on }]0, \pi[.$$

As $\tan x > x$ & $\cos x > 0$ on $]0, \pi[$.

$$\therefore f'(x) > 0 \quad \forall x \in]0, \pi[$$

$\therefore f$ is str. $\uparrow$ on $]0, \pi[$

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Let $f(x) = \frac{x}{\sin x}$

As f is str $\uparrow$ on $]0, \pi[$

$$\Rightarrow f(x) < f(\pi) \quad \forall x$$

$$\Rightarrow \frac{x}{\sin x} < \pi \quad \forall x$$

$$\therefore x < \pi \sin x \quad \forall x$$

Also At $x=0$ & $x=\pi$ $x = \frac{\pi}{2} \sin x$

Q13 (29.13) Let $h(x) = g(x) - f(x) \quad \forall x \in \mathbb{R}$

$$\therefore h'(x) = g'(x) - f'(x) \geq 0$$

$\therefore h$ is str $\uparrow$ on $\mathbb{R}$

$$\therefore h(x) \geq h(0) \quad \forall x \geq 0$$

$$g(x) - f(x) \geq g(0) - f(0)$$

$$\therefore g(x) \geq f(x) \quad \forall x \geq 0$$

Q14 (29.14) Let $g(x) = 2x - f(x)$ & $h(x) = f(x) - x$
Apply Same Method.

ex1) using M.V. thm s. that $e^x \geq 1+x$, $x \in \mathbb{R}$

Sol: Let $x=0$

Then $e^x = 1+x$ holds trivially.

$x > 0$

Let $f(x) = e^x$

$\therefore f$ is cont on $[0, x]$ & diff on $]0, x[$

then by M.V.Thm $\exists c \in]0, x[$ s.t

$$f'(c) = \frac{f(x) - f(0)}{x - 0}$$

$$e^c = \frac{e^x - e^0}{x - 0} \Rightarrow e^c x = e^x - 1$$

$$\therefore e^x = 1 + e^c x \quad \text{--- (1)}$$

Now $0 < c < x$

$\therefore c > 0$

$$\Rightarrow e^c > e^0$$

$$\Rightarrow x e^c > x$$

$$\Rightarrow 1 + x e^c > 1 + x$$

$$\Rightarrow e^x > 1 + x \quad \text{from (1)}$$

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ex2) Prove that $\frac{x-1}{x} < \log x < x-1$ $\forall x > 1$

Sol: Let $f(x) = \log x$ $\forall x > 0$

Let $x > 1$

$\therefore f$ is cont on $[1, x]$ and diff on $]1, x[$

by M.V. thm

$$\exists c \in]1, x[\text{ s.t } f'(c) = \frac{f(x) - f(1)}{x - 1}$$

$$\frac{1}{c} = \frac{\log x - \log 1}{x - 1}$$

$$\therefore \frac{1}{c} = \frac{\log x}{x - 1} \quad \text{--- (2)}$$

$$\log x = \frac{x-1}{c} \quad - (1)$$

$$\text{Now } c \in]1, x[\Rightarrow c > 1$$

$$\Rightarrow \frac{1}{c} < 1$$

$$\Rightarrow \frac{x-1}{c} < x-1 \quad (\text{as } x-1 > 0)$$

$$\therefore \text{from (1) gives } \log x < x-1 \quad - (2)$$

$$\text{Also } c < x \Rightarrow \frac{1}{c} > \frac{1}{x}$$

$$\Rightarrow \frac{x-1}{c} > \frac{x-1}{x}$$

$$\Rightarrow \log x > \frac{x-1}{x} \quad - (3)$$

from (2) & (3)

$$\frac{x-1}{x} < \log x < x-1 \quad \forall x > 1$$

Exercise 6.2

Q1 (a) $f(x) = x^2 - 3x + 5$

$$\therefore f'(x) = 2x - 3 \Rightarrow f'(x) = 0$$

$$\therefore x = \frac{3}{2}$$

$$x \leq \frac{3}{2} \Rightarrow 2x \leq 3 \Rightarrow f'(x) \leq 0 \therefore f \text{ is } \downarrow$$

$$x \geq \frac{3}{2} \Rightarrow 2x \geq 3 \Rightarrow f'(x) \geq 0 \therefore f \text{ is } \uparrow$$

The interval in which f is $\uparrow$ is $[\frac{3}{2}, \infty)$

" " " " f " $\downarrow$ is $(-\infty, \frac{3}{2}]$

Q2 (a) $h(x) = \sqrt{x} - 2\sqrt{x+2} \quad x > 0$

$$\therefore h'(x) = \frac{1}{2\sqrt{x}} - \frac{2}{2\sqrt{x+2}} \quad \forall x > 0$$

$$h'(x) = \frac{\sqrt{x+2} - 2\sqrt{x}}{2\sqrt{x}\sqrt{x+2}} = 0$$

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$$\sqrt{x+2} = 2\sqrt{x} \Rightarrow x+2 = 4x$$

$$\Rightarrow x = \frac{2}{3}$$

If $h''(x) < 0$

Then $h(x)$ is max at $x = \frac{2}{3}$

Q4 $f(x) = -\sum_{i=1}^n 2(a_i - x)$

$$\Rightarrow f'(x) = 0 \therefore \sum a_i - \sum x = 0$$

$$\Rightarrow nx = \sum_{i=1}^n a_i$$

$$\Rightarrow x = \frac{\sum a_i}{n}$$

$$\text{Now } f''(x) = 2 \sum_{i=1}^n 1 = 2n > 0$$

$\therefore$ point of relative minime $x = \frac{\sum a_i}{n}$

Q5 Define a function on $[-1; \infty)$ as

$$f(x) = x^{1/n} - (x-1)^{1/n} \quad \text{where } n \geq 2 \quad \forall n \in \mathbb{N}$$

$$\therefore f'(x) = \frac{1}{n} x^{\frac{1}{n}-1} - \frac{1}{n} (x-1)^{\frac{1}{n}-1}$$

$$= \frac{1}{n} \left[x^{\frac{1-n}{n}} - (x-1)^{\frac{1-n}{n}} \right] \quad \forall x > 1$$

$$= \frac{1}{n} \left[\frac{1}{x^{\frac{n-1}{n}}} - \frac{1}{(x-1)^{\frac{n-1}{n}}} \right] \quad \forall x > 1$$

$$\text{Now } x > 1 \Rightarrow x-1 > 0 \quad \& \quad x-1 < x$$

$$\Rightarrow \frac{1}{x-1} > \frac{1}{x}$$

$$\Rightarrow \frac{1}{(x-1)^{\frac{n-1}{n}}} > \frac{1}{x^{\frac{n-1}{n}}} \quad \forall n \geq 2$$

$$\therefore f'(x) < 0 \quad \forall x > 1$$

$$\Rightarrow f \text{ is str } \downarrow \text{ on } [1, \infty)$$

$$\Rightarrow f(x) < f(1) \quad \text{if } x > 1$$

$$\text{Now } a > b > 0 \Rightarrow \frac{a}{b} > 1 \Rightarrow f\left(\frac{a}{b}\right) < f(1)$$

$$\therefore \left(\frac{a}{b}\right)^{1/n} - \left(\frac{a}{b} - 1\right)^{1/n} < 1$$

$$\Rightarrow a^{1/n} - b^{1/n} < (a-b)^{1/n}$$

Q6 (do you self) v. Imp.

Q7 (" " ") v. Imp.

Q20 (v. Imp.)

a) by L.M.V. Thm

f is cont on $[0, 1]$, diff on $]0, 1[$

$$\exists c \in]0, 1[\text{ s.t. } f'(c) = \frac{f(1) - f(0)}{1 - 0}$$

$$\Rightarrow f'(c) = 1 - 0 = 1 \quad (1)$$

b) Rolle's Thm

f is cont on $[1, 2]$, diff on $]1, 2[$

$$\& f(1) = f(2) = 1$$

$$\exists c_2 \in]1, 2[\text{ s.t. } f'(c_2) = 0 \quad - (2)$$

c) Darboux thm

$$f'(c_1) = 1 \quad \& \quad f'(c_2) = 0$$

where $c_1 \in]0, 1[$ & $c_2 \in]1, 2[$

$\therefore \frac{1}{3}$ lies b/w $f'(c_1)$ & $f'(c_2)$

f is diff on $]0, 2[$ & $c_1, c_2 \in]0, 2[$

$$\text{s.t. } f'(c_2) < \frac{1}{3} < f'(c_1)$$

$$\therefore f'(c) = \frac{1}{3}$$

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Exercise 6.4

ex 1) use Taylor's thm with $n=2$, to approximate $(1+x)^{1/3}$ and hence find an approximation for $(1.3)^{1/3}$

Sol $f'(x) = \frac{1}{3} (1+x)^{-2/3}$

$$f''(x) = \frac{1}{3} \left(-\frac{2}{3}\right) (1+x)^{-5/3}$$

$$f'''(x) = -\frac{2}{9} \times -\frac{5}{3} (1+x)^{-8/3}$$

Applying Taylor's thm with Lagrange's Remainder After 3 terms, we get

Sol $f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(c)$, $0 < c < x$

$$(1+x)^{1/3} = 1 + x \cdot \frac{1}{3} + \frac{x^2}{2!} \left(-\frac{2}{9}\right) + \frac{x^3}{3!} \times \frac{10}{27} (1+c)^{-8/3}$$

$$= 1 + \frac{x}{3} - \frac{x^2}{9} + \frac{5}{81} x^3 (1+c)^{-8/3}$$

$$(1+x)^{1/3} - \left(1 + \frac{x}{3} - \frac{x^2}{9}\right) = \frac{5}{81} x^3 (1+c)^{-8/3}$$

So required Approximate of $(1+x)^{1/3}$ is given

by $1 + \frac{x}{3} - \frac{x^2}{9}$

Now $f(1.3) = ?$ Take $x = 0.3$

$$\Rightarrow (1.3)^{1/3} = 1 + \frac{0.3}{3} - \frac{0.3^2}{9} + \frac{5}{81} \frac{0.3^3}{10^2} (1+c)^{-8/3}$$

$$\Rightarrow (1.3)^{1/3} - \left(1 + \frac{1}{10} - \frac{1}{100}\right) = \frac{1}{600} (1+c)^{-8/3}$$

$$\Rightarrow (1.3)^{1/3} - \left(\frac{10+10-1}{100}\right) = \frac{1}{600} (1+c)^{-8/3}$$

$$\Rightarrow |(1.3)^{1/3} - 1.09| < \frac{1}{6} 10^{-2}$$

Approx. value. $\rightarrow$ degree of error.

ex2 Approximate the no. e with error less than 10^{-5} ?

sol $f(x) = e^x$

Apply Taylor's thm to $f(x)$ on $[0, x]$ with Lagrange's Remainder.

$$f(x) = f(0) + x f'(0) + \dots + \frac{x^{n-1}}{(n-1)!} f^{(n-1)}(0) + R_n$$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^{n-1}}{(n-1)!} + R_n$$

$$R_n = \frac{x^n}{n!} f^{(n)}(c) \quad , \quad 0 < c < x$$

$$= \frac{x^n}{n!} e^c$$

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Put $x=1$

$$\Rightarrow e = 1 + 1 + \frac{1}{2!} + \dots + \frac{1}{(n-1)!} + \frac{1}{n!} f^{(n)}(c) \quad , \quad 0 < c < 1$$

$$R^n(1) = \frac{f^{(n)}(c)}{n!} = \frac{e^c}{n!}$$

According to given $\frac{e^c}{n!} < 10^{-5}$

$$\Rightarrow \frac{e^c}{n!} < 10^{-5} \quad \text{if} \quad \frac{2}{n!} < 10^{-5} \quad e^c < 2$$

$$\text{i.e. } n! > 2 \times 10^5$$

$$\Rightarrow n! > 200000$$

Approximately $n=9$ as $9! > 200000$

$$\Rightarrow e \approx 1 + 1 + \frac{1}{2!} + \dots + \frac{1}{8!}$$

$$\approx 2.71828$$

Q4 $f(x) = (1+x)^{1/2} \quad x > 0.$

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Applying Taylor's thm. on $[0, x]$ with Lagrange's Remainder after n terms we get

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(c) \quad \text{--- (1) } 0 < c < x$$

Take $n=3$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(c)$$

$$(1+x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{x^3}{3!} \cdot \frac{3}{8}(1+c)^{-5/2}$$

$$(1+x)^{1/2} > 1 + \frac{1}{2}x - \frac{1}{8}x^2 \quad \text{--- (1)}$$

Take $n=2$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(c)$$

$$(1+x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2(1+c)$$

$$(1+x)^{1/2} \leq 1 + \frac{1}{2}x \quad \text{--- (2)}$$

from (1) & (2)

$$1 + \frac{1}{2}x - \frac{1}{8}x^2 \leq (1+x)^{1/2} \leq 1 + \frac{1}{2}x$$

Q5 Take $x=0.2$

$$1 + \frac{2}{10} \times \frac{1}{2} - \left(\frac{2}{10}\right)^2 \frac{1}{8} \leq (1.2)^{1/2} \leq 1 + \frac{1}{2} \left(\frac{2}{10}\right)$$

$$1 + \frac{19}{2 \times 10^2} < \sqrt{1.2} < 1 + \frac{1}{10}$$

$$\frac{219}{200} < \sqrt{1.2} < \frac{11}{10}$$

$$1.095 < \sqrt{1.2} < 1.1$$

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Q6 use Taylor's Thm with $n=2$ to obtain more Accuracy approx. for $\sqrt{1.2} \approx \sqrt{2}$ remainder after 3-terms, we get

$$(1+x)^{1/2} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{3!} \times \frac{3}{8} (1+c)^{-5/2}$$

Put $x=0.2$

$$(1.2)^{1/2} = \frac{219}{200} + \frac{(1+c)^{-5/2}}{2 \cdot 10^3}$$

$$\Rightarrow \left| (1.2)^{1/2} - \frac{219}{200} \right| = \left| \frac{(1+c)^{-5/2}}{2 \cdot 10^3} \right| \quad c > 0 \Rightarrow (1+c)^{-1} < 1$$

$$< \frac{1}{2 \cdot 10^3} \text{ Accuracy up to 3-degrees.}$$

$\therefore$ Approximate Value $\sqrt{1.2}$ is 1.091

Next Put $x=1$

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Q7 Same as Above

$$(1+x)^{1/3} = 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \frac{x^3}{3!} \times \frac{10}{27} (1+c)^{-8/3}$$

$$(1+x)^{1/3} - \left(1 + \frac{x}{3} - \frac{x^2}{9}\right) = \left| \frac{5x^3}{81} \right| (1+c)^{-8/3}$$

$$< \frac{5}{81} x^3$$

Q11 Let $f(x) = \log(1+x)$

Then, $f^n(x)$ exists $\forall n \in \mathbb{N}$ & $\forall x > 0$

$$\& f^n(x) = \frac{(-1)^{n-1}}{(1+x)^n} (n-1)!$$

Apply Taylor's Thm

$$\log(1+x) = x - \frac{x^2}{2!} + \dots + \frac{(-1)^{n-1}}{n} x^n + \frac{(-1)^n}{n+1} x^{n+1} \times \frac{1}{(1+c)^n}$$

$$(0 < c < x)$$

$$\Rightarrow \left| \log(1+c) - \left(c - \frac{c^2}{2} + \dots + \frac{(-1)^{n-1}}{n} c^n \right) \right|$$

$$= \left| \frac{(-1)^n}{n+1} c^{n+1} \right| \times \frac{1}{(1+c)^n}$$

Put $c = 0.5$

$$\left| \log(1.5) - \left(0.5 - \frac{0.5^2}{2} + \dots + \frac{(-1)^{n-1}}{n} (0.5)^n \right) \right| \leq \frac{0.5^{n+1}}{n+1}$$

$$< 0.001$$

$$\Rightarrow \left(\frac{5}{10} \right)^{n+1} \frac{1}{n+1} < \frac{1}{1000}$$

$$\Rightarrow n+1 > 1000 \left(\frac{1}{2} \right)^{n+1}$$

Sec-I

Q1 Show $\lim_{x \rightarrow 1} \frac{x^2 - x + 1}{x + 1} = \frac{1}{2}$

Q2 If $\lim_{x \rightarrow c} f(x) = L$ then $\lim_{x \rightarrow c} \sqrt{f(x)} = \sqrt{L}$

Q3 Show $\lim_{x \rightarrow 0^+} \sin \frac{1}{x}$ does not exist.

Q4 Proof of Sepⁿ char thm.

Sec-II

Q5 $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \notin \mathbb{Q} \end{cases}$

Show that for any $c \in \mathbb{R}$, $\lim_{x \rightarrow c} f(x)$ does not exist

Q6 Define Signum funⁿ $\text{sgn}(x)$, and show that $\lim_{x \rightarrow 0} \text{sgn}(x)$ does not exist.

Sol: $\text{sgn} : \mathbb{R} \rightarrow \mathbb{R}$ defined as:

$$\text{sgn}(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0 \end{cases}$$

Let $f(x) = \text{sgn}(x)$

If possible, Suppose $\lim_{x \rightarrow 0} f(x)$ exist = L

Consider the Sepⁿ $\langle x_n \rangle = \langle \frac{(-1)^n}{n} \rangle$

clearly $\langle x_n \rangle \rightarrow 0$

$\therefore$ by Sepⁿ char thm. $\langle f(x_n) \rangle \rightarrow L$

Now $\langle x_n \rangle = \langle -1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, \dots \rangle$

$\langle f(x_n) \rangle = \langle -1, 1, -1, 1, -1, 1, \dots \rangle$

Which does not conv.

$\therefore$ Contradict ϵ .

$\lim_{x \rightarrow 0} \sin(x)$ does not exist.

Q7 Show $f(x) = \sin x$ is cont on $\mathbb{R}$.

HINT $[|x-c| < \delta \Rightarrow |\sin x - \sin c| < \epsilon]$

Consider $|\sin x - \sin c|$

$$= \left| 2 \cos \frac{x+c}{2} \sin \frac{x-c}{2} \right|$$

$$\leq 2 \left| \frac{\sin(x-c)}{2} \right|$$

$$\leq |x-c| < \delta$$

$$\text{Let } \delta = \epsilon.$$

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Q8 Use Location of Root thm to prove the function $f(x) = 2 \log x - \sqrt{x} - 2$ has a root in $[1, 2]$

Soln:- f is cont on $[1, 2]$

$$f(1) = 2 \log 1 - \sqrt{1} - 2 = 2 \cdot 0 - 1 - 2 < 0$$

$$f(2) = 2 \log 2 - \sqrt{2} - 2 = \log 4 + \sqrt{2} - 2$$

$$> \log e + \sqrt{2} - 2$$

$$= 2.7 + 1.4 - 2 > 0$$

$\therefore$ by L. of Root thm

$$\exists c \in]1, 2[\text{ s.t. } f(c) = 0$$

$$2 \log c - \sqrt{c} - 2 = 0$$

$\therefore c$ is a root of the eqn.

Q9 Let $f: [0,1] \rightarrow \mathbb{R}$ be cont. & take only rational values. Show f is constant on $[0,1]$

Solⁿ If possible, Suppose f is not const $[0,1]$
 $\therefore \exists a, b \in [0,1]$ s.t. $f(a) \neq f(b)$
 WLOG, let $f(a) < f(b)$

$\therefore \exists$ a rational k s.t. $f(a) < k < f(b)$

by B.I.V.Thm. $\exists c \in [0,1]$ s.t. $f(c) = k$. (Cont. f takes $\mathbb{Q}$ values)

$\therefore f$ must be const on $[0,1]$

~~which con~~

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Q10 Show that $f(x) = \frac{5x}{2x-1}$ is unif cont on $[1, \infty)$

Solⁿ Consider $\left| \frac{5x}{2x-1} - \frac{5y}{2y-1} \right| = \frac{5}{(2x-1)(2y-1)} |x-y|$ — (1)

$$x \geq 1, y \geq 1$$

$$\Rightarrow 2x \geq 2, 2y \geq 2$$

$$\Rightarrow 2x-1 \geq 1, 2y-1 \geq 1$$

$$\Rightarrow \frac{1}{2x-1} \leq 1, \frac{1}{2y-1} \leq 1$$

$$\delta = \frac{5}{5}$$

(OR) Show $f(x) = e^{ix} \sin x - e^{ix} \cos 3x$ is unif cont on $[0, \pi]$

Solⁿ on $[0, \pi]$ cont $\Rightarrow$ unif. cont.

$\therefore$ As f is cont on $[0, \pi]$

$\therefore$ unif cont on $[0, \pi]$

Q10 Show $-x \leq \sin x \leq x \quad \forall x \geq 0$

Let $f(t) = \sin t$ on $[0, x]$

f is cont on $[0, x]$

" " diff " $]0, x[$

by L.M.V. thm

$$\exists c \in]0, x[\text{ s.t. } f'(c) = \frac{f(x) - f(0)}{x - 0}$$

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$$x f'(c) = f(x) - f(0)$$

$$x \cos c = \sin x \quad \text{--- (1)}$$

$$-1 \leq \cos c \leq 1$$

$$\Rightarrow -x \leq x \cos c \leq x$$

$$\Rightarrow -x \leq \sin x \leq x \quad \text{from (1)}$$

OR Show $x < \tan x \quad \forall x \in]0, \pi/2[$

Let $f(x) = \tan x - x$

$$\therefore f'(x) = \sec^2 x - 1 = \tan^2 x > 0$$

$\therefore f$ is str $\uparrow$ on $[0, \pi/2]$

$$\therefore f(x) > f(0)$$

$$\Rightarrow \tan x - x > 0$$

$$\Rightarrow \tan x > x$$

OR Show $\frac{x}{\sin x} \leq \pi/2 \quad \forall x \in [0, \pi/2]$

Let $f(x) = \frac{x}{\sin x}$

$$\therefore f'(x) = \frac{\cos x}{\sin^2 x} (\tan x - x) > 0$$

(5)

$\therefore f$ is str $\uparrow$ on $]0, \pi/2[$

$$\therefore g(u) < g(\pi/2)$$

$$\text{i.e. } \frac{u}{\sin u} < \frac{\pi/2}{\sin \pi/2}$$

$$\Rightarrow \frac{u}{\sin u} < \pi/2 \quad \forall u \in]0, \pi/2[.$$

Q11 State & prove Cauchy M.V. thm.

Q12 Let $f(u) = \begin{cases} u^2 \sin \frac{1}{u} & u \neq 0 \\ 0 & u = 0 \end{cases}$

Show that f' exist but is not cont on $\mathbb{R}$

Solⁿ
(HINT) Consider $\lim_{u \rightarrow 0} \frac{u^2 \sin \frac{1}{u} - 0}{u - 0} = \lim_{u \rightarrow 0} u \sin \frac{1}{u}$

$= 0$
(If $\lim_{u \rightarrow 0} f(u) = 0$ & $g(u)$ is bdd then $\lim_{u \rightarrow 0} f(u)g(u) = 0$)

$\therefore f$ is diff at 0

$$f'(u) = \begin{cases} 2u \sin \frac{1}{u} - \cos \frac{1}{u} & u \neq 0 \\ 0 & u = 0 \end{cases}$$

Now test for cont of $f'(u)$.

($\cos \frac{1}{u}$ limit does not exist)

Sec 4

Q13 Define Inverse function thm.
 & use it to verify

① $f(x) = x^5 + 4x^3 + 3$ at $x=1$ on $\mathbb{R}$

Sol f is cont on $\mathbb{R}$

$$f'(x) = 5x^4 + 12x^2 > 0$$

$\therefore f$ is str $\uparrow$ on $\mathbb{R}$.

$\therefore$ its inverse $g = f^{-1}$ exists.

As f is diff at 1 & $f'(1) = 9 \neq 0$

$\therefore$ by Inverse thm.

g is diff at $f(1) = 8$

$$\& g'(8) = \frac{1}{f'(1)} = \frac{1}{9}$$

OR $f(x) = x^3 + 2x + 1$ at $x=0, 1, -1$

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Q14 Let $f(x) = x^2$ $x \geq 0$
 $= 0$ $x < 0$

1) sketch graph of f

2) Show f is diff at 0

3) calculate f' & sketch its graph

4) Is f' is cont on $\mathbb{R}$? diff on $\mathbb{R}$?

Ans

1) $\checkmark$

2) $Lf'(0) = Rf'(0) = 0$

④ $\checkmark$ Not diff

$Lf'(0) \neq Rf'(0)$

3) $f'(x) = 2x$ $x \geq 0$
 0 $x < 0$

$\checkmark$ but cont

Sec 4Q15 If $x \in [0, 1]$ & $n \in \mathbb{N}$ prove

$$\left| \log(1+x) - \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots - (-1)^{n-1} \left(\frac{x^n}{n} \right) \right| < \frac{x^{n+1}}{n+1}$$

use this to Approx $\log 1.5$ with error less than 0.01

Q16 Define convex fun. & its thm.

(A convex fun. need not be diff)

 $f(x) = |x|$ is not convex on $\mathbb{R}$.

$$f(x) = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

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$$L f'(0) = \lim_{x \rightarrow 0^-} \frac{-x - 0}{x - 0} = -1$$

$$R f'(0) = \lim_{x \rightarrow 0^+} \frac{x - 0}{x - 0} = 1$$

$$\therefore L f'(0) \neq R f'(0)$$

 $\therefore f$ is not convex.Q17 Taylor's expansion for $\sin 2x$

Q18 State & prove Taylor's thm.

shows that

$$g'(y_0) = \frac{1}{yx_0^{n-1}} = \frac{1}{(y_0^{n-1})^n} = \frac{1}{y_0^{n-1}}.$$

This shows that the function g is differentiable for $y \neq 0$ and that the rule for differentiating x^n holds for exponents of the form $1/n$. See also Exercise 29.15.

Theorem 29.9 applies to the various inverse functions encountered in calculus. We give one example.

Example 3

The function $f(x) = \sin x$ is one-to-one on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and it is traditional to use the inverse g of f restricted to this domain; g is usually denoted $\sin^{-1}$ or $\arcsin$. Note that $\text{dom}(g) = [-1, 1]$. For $y_0 = \sin x_0$ in $(-1, 1)$ where $x_0 \in (-\frac{\pi}{2}, \frac{\pi}{2})$, Theorem 29.9 shows that $g'(y_0) = \frac{1}{\cos x_0}$. Since $1 = \sin^2 x_0 + \cos^2 x_0 = y_0^2 + \cos^2 x_0$ and $\cos x_0 > 0$, we may write

$$g'(y_0) = \frac{1}{\sqrt{1-y_0^2}}$$

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Exercises

29.1. Determine whether the conclusion of the Mean Value Theorem holds for the following functions on the specified intervals. If the conclusion holds, give an example of a point x satisfying (1) of Theorem 29.3. If the conclusion fails, state which hypotheses of the Mean Value Theorem fail.

- (a) x^2 on $[-1, 2]$,
- (c) $|x|$ on $[-1, 2]$,
- (e) $\frac{1}{x}$ on $[1, 3]$,
- (b) $\sin x$ on $[0, \pi]$,
- (d) $\frac{1}{x}$ on $[-1, 1]$,
- (f) $\text{sgn}(x)$ on $[-2, 2]$.

The function sgn is defined in Exercise 17.10.

29.2. Prove that $|\cos x - \cos y| \leq |x - y|$ for all $x, y \in \mathbb{R}$.

29.3. Suppose that f is differentiable on $\mathbb{R}$ and that $f(0) = 0$, $f(1) = 1$ and $f(2) = 1$.

- (a) Show that $f'(x) = \frac{1}{2}$ for some $x \in (0, 2)$.
- (b) Show that $f'(x) = \frac{1}{2}$ for some $x \in (0, 2)$.

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29.4. Let f and g be differentiable functions on an open interval I . Suppose that a, b in I satisfy $a < b$ and $f(a) = f(b) = 0$. Show that $f'(x) + f(x)g'(x) = 0$ for some $x \in (a, b)$. Hint: Consider $h(x) = f(x)e^{g(x)}$.

29.5. Let f be defined on $\mathbb{R}$, and suppose that $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Prove that f is a constant function.

29.6. Give the equation of the straight line used in the proof of the Mean Value Theorem 29.3.

29.7. (a) Suppose that f is twice differentiable on an open interval I and that $f''(x) = 0$ for all $x \in I$. Show that f has the form $f(x) = ax + b$ for suitable constants a and b .

(b) Suppose f is three times differentiable on an open interval I and that $f''' = 0$ on I . What form does f have? Prove your claim.

29.8. Prove (ii)-(iv) of Corollary 29.7.

29.9. Show that $ex \leq e^x$ for all $x \in \mathbb{R}$.

29.10. Let $f(x) = x^2 \sin(\frac{1}{x}) + \frac{1}{2}$ for $x \neq 0$ and $f(0) = 0$.

- (a) Show that $f'(0) > 0$; see Exercise 28.4.
- (b) Show that f is not increasing on any open interval containing 0.
- (c) Compare this example with Corollary 29.7(i).

29.11. Show that $\sin x \leq x$ for all $x \geq 0$. Hint: Show that $f(x) = x - \sin x$ is increasing on $[0, \infty)$.

29.12. (a) Show that $x < \tan x$ for all $x \in (0, \frac{\pi}{2})$.

- (b) Show that $\frac{x}{\sin x}$ is a strictly increasing function on $(0, \frac{\pi}{2})$.
- (c) Show that $x \leq \frac{1}{2} \sin x$ for $x \in [0, \frac{\pi}{2}]$.

29.13. Prove that if f and g are differentiable on $\mathbb{R}$, if $f(0) = g(0)$ and if $f'(x) \leq g'(x)$ for all $x \in \mathbb{R}$, then $f(x) \leq g(x)$ for $x \geq 0$.

29.14. Suppose that f is differentiable on $\mathbb{R}$, that $1 \leq f'(x) \leq 2$ for $x \in \mathbb{R}$, and that $f(0) = 0$. Prove that $x \leq f(x) \leq 2x$ for all $x \geq 0$.

29.15. Let r be a nonzero rational number $\frac{m}{n}$ where n is a positive integer, m is any nonzero integer, and m and n have no common factors. Let $h(x) = x^r$ where $\text{dom}(h) = [0, \infty)$ if n is even and $m > 0$.

